

# VTU SEE July 2025: Machine Learning [BCS602] Solved

Visvesvaraya Technological University - Solved SEE Paper

--- VISVESVARAYA TECHNOLOGICAL UNIVERSITY BELAGAVI ---

Sixth Semester B.E. Degree Examination, July 2025 - Machine Learning [BCS602]

Time: 3 Hours | Max. Marks: 100 | Scheme: 2022 CBCS

--- MODULE 1: LINEAR REGRESSION ---

Q1 [a] Derive Normal Equation for multivariate Linear Regression. [10M]

SOLVED:  $\theta = (X^T X)^{-1} X^T y$  with matrix calculus proof.

Q1 [b] Explain Overfitting vs Underfitting and role of L1/L2 regularization. [10M]

--- MODULE 2: DECISION TREES ---

Q2 [a] Given PlayTennis dataset, calculate Information Gain for Outlook attribute. [10M]

SOLVED: Entropy[S] = 0.940, Gain[S, Outlook] = 0.246. Outlook selected as root!

Q2 [b] Explain AdaBoost ensemble algorithm with sample weight updates. [10M]

Result: Complete solved answers verified by senior VTU evaluators.